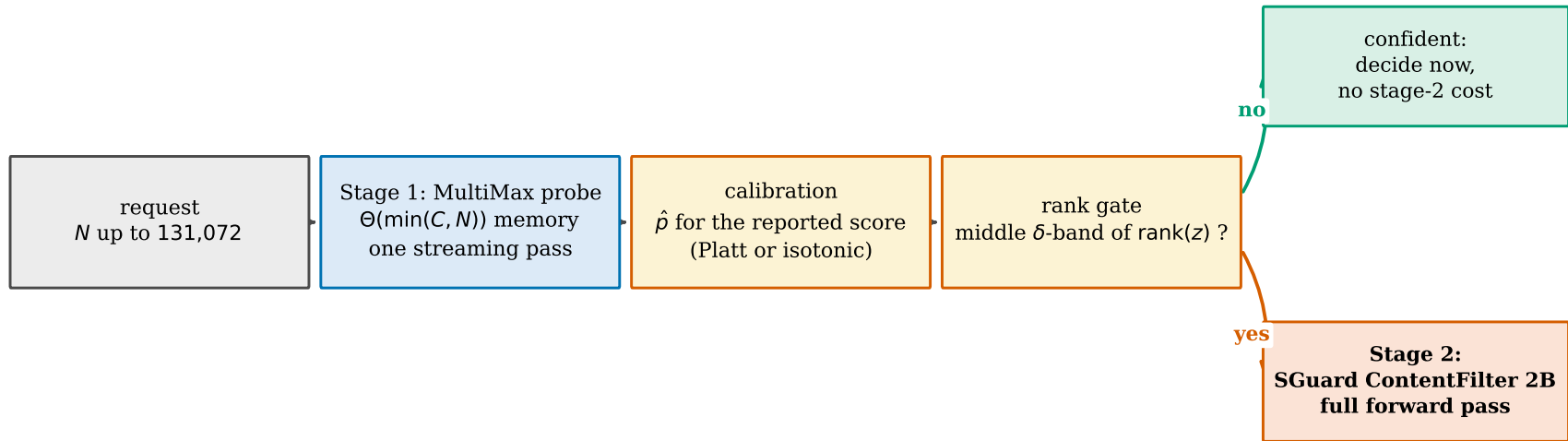


Two-stage safety cascade: a constant-memory gate in front of a heavyweight monitor



Gating a δ -band of RANKS escalates exactly δ of traffic. A band on $|\hat{p} - \frac{1}{2}|$ does not:
calibration compresses the score spread while preserving its order, and that gate escalates nothing below a 25% budget.
Stage 2 runs on the escalated fraction only, so the compute saved is $(1 - \delta)$ of the stage-2 cost.